8-Queens Problem

1. Aim

To implement the **8-Queens problem** using an Artificial Intelligence search technique to place eight queens on an 8×8 chessboard such that no two queens attack each other.

2. Problem Statement

The 8-Queens problem is a classic Artificial Intelligence problem in which **eight queens must be placed on an 8×8 chessboard**. The queens must be positioned so that no two queens share the same row, column, or diagonal. The objective is to find a valid arrangement that satisfies all these constraints.

3. Solution

The problem can be solved using the **backtracking technique**. Queens are placed one row at a time, and each position is checked to ensure that the new queen does not conflict with any previously placed queen. If a safe position is not available, the algorithm backtracks to the previous row and tries another position. This process continues until all eight queens are successfully placed.

4. Algorithm

1. Start with an empty 8×8 chessboard.
2. Place a queen in the first row.

3. Check whether the selected position is safe.
4. A position is safe if no other queen exists in the same column or diagonal.
5. If the position is safe, place the queen and move to the next row.
6. If no safe position is available, backtrack to the previous row.
7. Move the previous queen to another possible position.
8. Repeat the process until all eight queens are placed.
9. Display the final chessboard arrangement.

5. Program

```
def is_safe(board, row, col):
    for i in range(row):
        # Same column
        if board[i] == col:
            return False

        # Same diagonal
        if abs(board[i] - col) == abs(i - row):
            return False
```

```
    return True


def solve_queens(board, row):
    if row == 8:
        return True

    for col in range(8):
        if is_safe(board, row, col):
            board[row] = col

            if solve_queens(board, row + 1):
                return True

            # Backtrack
            board[row] = -1

    return False


board = [-1] * 8
```

```
if solve_queens(board, 0):
    print("Solution Found:\n")

    for row in range(8):
        for col in range(8):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No solution exists.")
```

6. Result

The 8-Queens problem was successfully implemented using the **backtracking algorithm**. The program checks the possible positions of each queen and rejects positions that cause conflicts with previously placed queens.

Output:

Solution Found:

Q . . . . . .

. . . . Q . . .

. . . . Q

. . . . . Q . .

. . Q . . . . .

. . . . . Q .

. Q . . . . . .

. . . Q . . . .

Thus, eight queens were successfully placed on the chessboard such that **no two queens attack each other**, satisfying the constraints of the 8-Queens problem.